

$$m = 10.0 \text{ kg}$$

$$K = 2250 \text{ N/m}$$

$$h = 3.0 \text{ m}$$

$$L = 6.0 \text{ m}$$

$$\Delta E = W$$

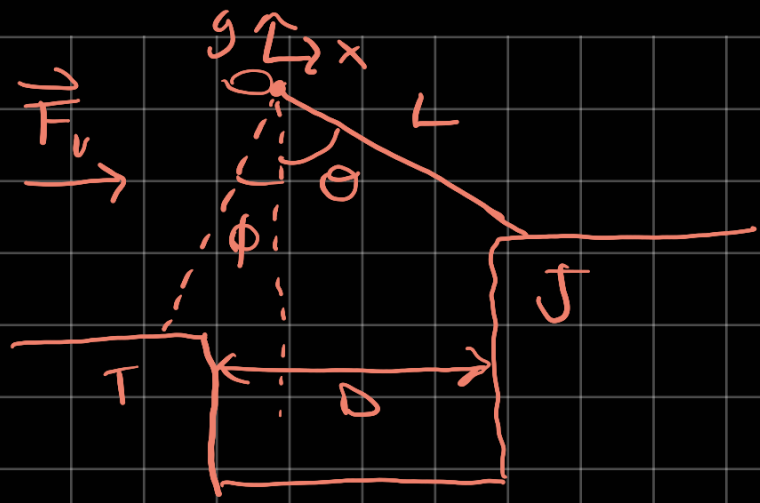
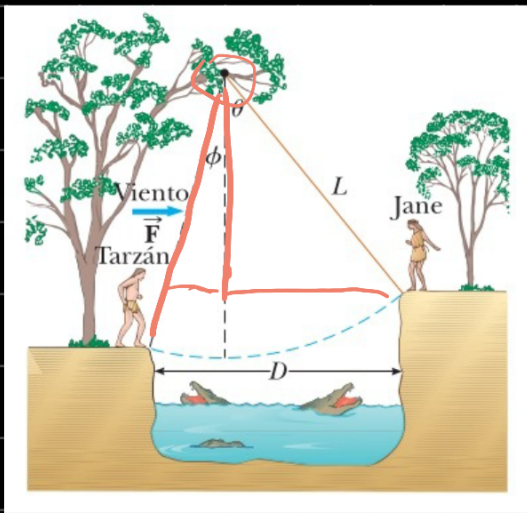
$$\Delta E = \Delta K + \Delta U_g = -f_k L$$

$$\Rightarrow K_f - K_i + U_{gf} - U_{gi} = -f_k L$$

$$\Rightarrow \frac{1}{2} K x^2 - mgh = -\mu mgL$$

$$\Rightarrow \mu mgL = mgh - \frac{1}{2} K x^2$$

$$\Rightarrow \mu = \frac{mgh - \frac{1}{2} K x^2}{mgL} = 0.328$$



$$m_s = 50.0 \text{ kg} \quad m_r = 80. \text{ kg}$$

$$\theta = 50.0^\circ \quad L = 40.0 \text{ m} \quad D = 30.0 \text{ m}$$

$$D = L \sin \theta + L \sin \phi$$

$$\Rightarrow L \sin \phi = D - L \sin \theta$$

$$\sin \phi = \frac{D - L \sin \theta}{L}$$

$$\phi = \sin^{-1} \left(\frac{D - L \sin \theta}{L} \right) = 28.9^\circ$$

$$\Delta E = W \Rightarrow E_f - E_i = W$$

$$E_f = W + E_i$$

$$\Rightarrow K_f + U_{g_f} = -F_v D + K_i + U_{g_i}$$

$$-mgL \cos \phi = -F_v D + \frac{1}{2} m v_i^2 - mgL \cos \theta$$

$$\Rightarrow \frac{1}{2} m v_i^2 = mgL (\cos \theta - \cos \phi) + F_v D$$

$$v_i = \sqrt{\frac{2mgL (\cos \theta - \cos \phi) + F_v D}{m}} = 6.13 \text{ m/s}$$

$$M = m_s + m_T = 130 \text{ kg}$$

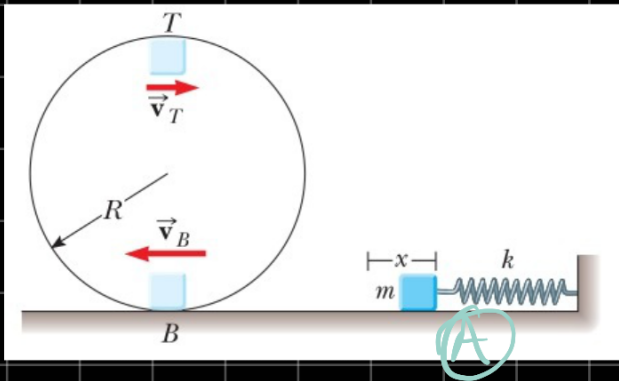
$$\Delta E = W \Rightarrow \cancel{K_f} + U_{pf} = W + K_i + U_{gi}$$

$$= -MgL \cos \theta = F_v \Delta + \frac{1}{2} M v_i^2 - MgL \cos \phi$$

$$\Rightarrow \frac{1}{2} M v_i^2 = MgL (\cos \phi - \cos \theta) - F_v \Delta$$

$$\Rightarrow v_i = \sqrt{\frac{2MgL (\cos \phi - \cos \theta) - F_v \Delta}{M}}$$

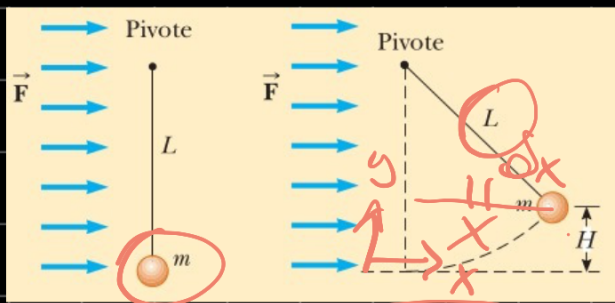
$$= 9.89 \text{ m/s}$$



$$W_{AB} = -26.25 \text{ J}$$

$$W_{BT} = 15.7 \text{ J}$$

$$N = 15.7 \text{ N}$$



Pitágoras

$$x^2 = L^2 - (L-H)^2$$

$$x = \sqrt{L^2 - (L-H)^2}$$

$$x^2 + (L-H)^2 = L^2 \quad \text{Hip}$$

Trabajo del viento $\Rightarrow W = \int \vec{F} \cdot d\vec{s}$

$$\vec{F} = F \hat{i}, \quad d\vec{s} = dx \hat{i}$$

$$\Rightarrow W = \int F dx (\hat{i} \cdot \hat{i}) = \int F dx = F \int dx$$

$$\int dx = x \Rightarrow W = F \sqrt{2LH - H^2}$$

$$\Delta E = W \Rightarrow E_f - \cancel{E_i} = W$$

$\hookrightarrow$ repose

$$\cancel{K_f} + U_{gf} = F \sqrt{2LH - H^2}$$

$$\Rightarrow mgh = F \sqrt{2LH - H^2} / (1)^2$$

$$\Rightarrow m^2 g^2 H^2 = F^2 (2LH - H^2)$$

$$\Rightarrow m^2 g^2 H = 2LF^2 - F^2 H$$

$$\Rightarrow H(m^2 g^2 + F^2) = 2LF^2$$

$$\Rightarrow H = \frac{2L \cancel{F^2}}{m^2 g^2 + \cancel{F^2}} = \frac{2L}{\frac{m^2 g^2}{F^2} + 1}$$

$$\Rightarrow H = \frac{2L}{1 + \frac{m^2 g^2}{F^2}}$$

(b) H wenn $F \rightarrow 0 \wedge F \rightarrow \infty$
 $F \rightarrow 0 \quad H = 0$

$$H = \frac{2L}{1 + \left(\frac{mg}{F}\right)^2} = 2L \left(1 + \left(\frac{mg}{F}\right)^2\right)^{-1} \quad \textcircled{-1}$$

Serie Binomial

$$(1+x)^{\textcircled{m}} = \sum_{n=0}^{\infty} \binom{m}{n} x^n =$$

$$= 1 + mx + \frac{m(m-1)}{2!} x^2 + \dots$$

$$m = -1, \quad x = \left(\frac{mg}{F}\right)^2$$

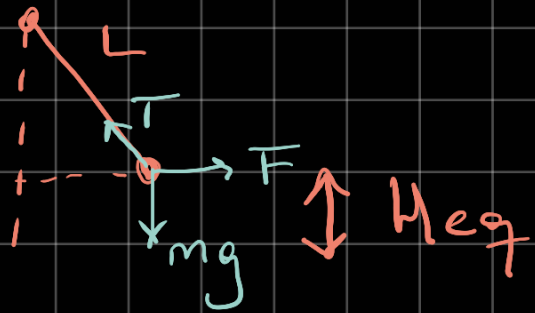
$$H = 2L \left(1 + \left(\frac{mg}{F}\right)^2\right)^{-1} \approx 2L \left(1 - \left(\frac{mg}{F}\right)^2\right) \dots$$

$$\frac{m(m-1)}{2!} = \frac{-1(-1-1)}{2} = \frac{2}{2} = 1$$

$$H \approx 2L \left(1 - \left(\frac{mg}{F} \right)^2 + \left(\frac{mg}{F} \right)^4 + \dots \right)$$

$$\boxed{H \approx 2L} \quad F \rightarrow \infty$$

c) $\Rightarrow$



$$H_{eq} = L - L \cos \theta = L \left(1 - \underbrace{\cos \theta}_? \right)$$

Fuerzas

$$\sum F_x = 0 \Rightarrow F - T \sin \theta = 0 \quad | :1$$

$$1) \sum F_y = 0 \Rightarrow T \cos \theta - mg = 0$$

$$T \cos \theta = mg \Rightarrow \cos \theta = \frac{mg}{T}$$

$$F = T \sin \theta$$

$$+ mg = T \cos \theta$$

$$F + mg = T \sin \theta + T \cos \theta$$

$$F^2 + m^2 g^2 = T^2 (\underbrace{\sin^2 \theta + \cos^2 \theta}_1)$$

$$F^2 + m^2 g^2 = T^2$$

$$T = \sqrt{F^2 + m^2 g^2}$$

$$H_{eq} = 2 \left(1 - \frac{m g x}{\sqrt{F^2 + m^2 g^2}} \right)$$

$$= 2 \left(1 - \frac{1}{\sqrt{1 + \frac{F^2}{m^2 g^2}}} \right)$$

